

JIawei CHEN

Crystal Structure Prediction (CSP) / Machine Learning (ML) / First-Principles Calculations

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 Guangdong University of Technology  M.S. in Materials Physics and Chemistry

Education

2027.06	School of Physics and Optoelectronic Engineering, Guangdong University of Technology
2024.09	M.S. Candidate in Materials Physics and Chemistry
2024.06	School of Chemical Engineering and Light Industry, Guangdong University of Technology
2020.09	B.S. in Applied Chemistry (Exemption from graduate entrance examination)

Honors & Awards

- ▶ Graduate Elite Talent Program, Guangdong University of Technology
- ▶ First-Class Graduate Scholarship (2024, 2024–2025), Guangdong University of Technology
- ▶ Third Prize, South China Undergraduate Physics Experiment Competition (2022)
- ▶ Second Prize, Guangdong Division of the National Mathematical Contest in Modeling (2022)
- ▶ Outstanding Student Scholarship (2020–2023), Guangdong University of Technology

Research Experience

2026.05	USPEX-Analyzer: Browser-based analysis tool for USPEX V10.5
2026.04	▶ Developed a browser-based post-processing tool for USPEX outputs, supporting interactive convex-hull/Pareto/genealogy visualization, multi-condition filtering, and export
present	CSP of Hydrogen-Based Superconductors via Multi-Objective Optimization
2026.01	▶ Embedded Interpretable Prediction Model into USPEX's multi-objective optimization to co-optimize thermodynamic stability and superconducting properties
	▶ Establishing a closed-loop workflow of search–screen–validate–update
present	CSP Driven by Universal Machine Learning Potentials
2025.01	▶ Integrated universal ML potentials into USPEX for faster CSP
	▶ Fine-tuned MatterSim to improve prediction accuracy under high pressure
2026.01	High-Throughput Screening of Conventional Superconductors
2025.01	▶ Combined structure generators with MatterSim for high-throughput screening of hydrogen-based superconductors
	▶ Accelerated substitution screening of layered boride superconductors using MatterSim
2026.01	Interpretable Prediction Model for T_c of Hydrogen-Based Superconductors
2024.09	▶ Collected, preprocessed datasets and performed feature engineering
	▶ Constructed interpretable models using SISSO
	▶ Revealed relationships between pressure, electronic structure, and superconductivity
2024	Defect Engineering for Hydrogen Evolution Reaction in 2D Materials
2022	▶ Investigated the effects of defects on electronic structure and HER activity of HfX_2 ($X = \text{S, Se, Te}$)

Skills

Programming	Python, Bash, LaTeX
Software	VASP, Quantum ESPRESSO, USPEX, SISSO, Lammps

Publications

- ▶ **Chen J**, Peng J, Liang Y, et al. Interpretable descriptors enable prediction of hydrogen-based superconductors at moderate pressures. *Materials Today Physics*, 2026, 63: 102073.
- ▶ Liang Y*, **Chen J***, Huang Y, et al. High-Throughput Screening of Bulk Boride Superconductors with Honeycomb-Ruby Frameworks. *Inorganic Chemistry*, 2026, 65(10): 5731–5739.
- ▶ **Chen J***, Zhang R*, Luo J, et al. Activating HfX_2 ($X = \text{S, Se and Te}$) for the hydrogen evolution reaction by introducing defects: a first-principles study. *Physical Chemistry Chemical Physics*, 2023, 25(38): 26043–26048.